EE3005 Communication Systems - Assignment 4

Department of Electrical Engineering, IIT Madras

March 2026

Topics covered: Effective noise bandwidth, Matched filter, SNR, Noise factor/figure, Friis formula, SNR of AM/FM, Analog communications link budget/analysis, Passband/baseband sampling, Uniform and non-uniform Quantization

Instructions: Submit your solutions in one PDF file with all step-by-step answers, plots and outputs written/printed in that file.

---

1. A communication system has a power gain of 20 dB and operates over a bandwidth of 1 MHz. The input signal power is 2 μW. The input noise power spectral density is $2 \times 10^{-15}\, W/Hz$. Determine the following.

   a) Input noise power.

   b) Output signal power.

   c) Output noise power.

   d) Input and output signal-to-noise ratio.

   e) Noise Factor.

2. In a communication system, the following signal and noise powers are measured at the input and output of a receiver. The input signal power is $5\,\mu W$ and the input noise power is $0.05\,\mu W$. At the output of the receiver, the signal power is $200\,\mu W$ and the output noise power is $8\,\mu W$.

   a) The Noise Factor F of the receiver.

   b) The Noise Figure NF of the receiver in dB.

3. A communication receiver consists of three cascaded amplifier stages with the following parameters:

Stage	Gain (dB)	Noise Figure (dB)
1	12	3
2	10	6
3	8	7

Determine the following:

a) Using Friis formula, determine the overall noise factor of the cascaded system.
b) Calculate the overall noise figure of the receiver in dB.

4. State whether the following statements are True or False. Provide a clear justification for your answer.

a) In a communication receiver, if the input signal-to-noise ratio is SNR_{in} and the noise figure of the receiver is $NF = 6\,$dB, then the output signal-to-noise ratio is always equal to $SNR_{in} - 6\,$dB.
b) If the noise power spectral density N_0 is constant over the receiver bandwidth B, the input noise power is $N_{in} = N_0 B$. If the bandwidth of the receiver is doubled while the signal power remains unchanged, the input SNR will decrease by a factor of two.

5. Consider the signal
$$s(t) = I_{[0,2]}(t) - 2I_{[1,3]}(t).$$

(a) Find and sketch the impulse response $s_{mf}(t)$ of the matched filter for s.
(b) Find and sketch the output when $s(t)$ is passed through its matched filter.
(c) Suppose that, instead of the matched filter, all we have available is a filter with impulse response
$$h(t) = I_{[0,1]}(t).$$
For an arbitrary input signal $x(t)$, show how
$$z(t) = (x * s_{mf})(t)$$
can be synthesized from
$$y(t) = (x * h)(t).$$

6. A satellite in geosynchronous orbit (36,000 km above the earth's surface) radiates 100 W of power (20 dBW). The transmitting antenna has a gain of 18 dB, so that the EIRP = 38 dBW. The earth station employs a 3-meter parabolic antenna and the downlink is transmitting at a frequency of 4 GHz. Determine the received power. Note: EIRP is the product of P_T and G_T. Take η (illumination efficiency factor) to be 0.5 and area of parabolic antenna to be A= $\frac{\pi D^2}{4}$

Hint: Use the formula
$$G_R = \frac{\eta A\, 4\pi}{\lambda^2}.$$

7. A baseband signal $x(t)$ has spectrum
$$X(f) = \begin{cases} 1 - |f|/10, & |f| \leq 10 \text{ kHz} \\ 0, & \text{otherwise} \end{cases}$$

The signal is sampled with sampling frequency $f_s = 15$ kHz.

1. Determine whether aliasing occurs.
2. Determine the frequency range over which spectral overlap occurs.
3. If an ideal low-pass reconstruction filter with cutoff 10 kHz is used, determine whether perfect reconstruction is possible.

8. A bandpass signal occupies the frequency band

$$120 \leq |f| \leq 150 \text{ kHz}.$$

1. Determine the bandwidth of the signal.
2. Using the bandpass sampling theorem, determine all possible sampling frequency ranges for $k = 2$.
3. Determine whether $f_s = 80$ kHz satisfies the bandpass sampling condition.

9. A bandpass signal originally occupies the frequency band

$$f_L \leq |f| \leq f_H.$$

After sampling at frequency $f_s = 40$ kHz, the aliased spectrum appears in the band

$$5 \leq |f| \leq 15 \text{ kHz}.$$

1. Determine the bandwidth of the signal.
2. Determine the original frequency band of the signal.
   Hint: Aliased frequencies after sampling occur at

$$f_{alias} = |f - kf_s|.$$

3. Determine the carrier frequency of the signal.

10. A modulated signal

$$x(t) = m(t)\cos(2\pi f_c t)$$

has message bandwidth $B = 8$ kHz and carrier frequency $f_c = 60$ kHz.

1. Determine the frequency band occupied by the signal.
2. Determine the Nyquist sampling rate.
3. Determine the minimum sampling rate using bandpass sampling.
4. If $f_s = 25$ kHz, determine the location of the aliased spectrum.

11. Assume that the input to a uniform quantiser is symmetric and lies in $[-V, V]$. Define the loading factor

$$\alpha = \frac{V}{\sigma},$$

where σ^2 is the input signal power. Express the SNR (in dB) of the quantiser in terms of the number of bits per sample R and α. Use the usual large-N approximation where $N = 2^R$. Also compute the SNR for a sine wave with amplitude V.

12. A non-uniform quantiser operating between $-3V$ and $+3V$ has six segments. Each segment has 16 uniformly spaced quantisation levels and the step size doubles in each segment starting from the origin. The input has a uniform pdf on $[-3V, 3V]$.

    (a) Determine the number of bits needed to represent the quantised samples.

    (b) Determine the signal-to-noise ratio.

    (c) Determine the pdf of the quantisation noise.

13. An 8-bit ADC is used for sampling and quantising a signal with a dynamic range of $[-1, 1]$ V.

    (i) Determine the quantisation noise variance.

    (ii) For a sinusoidal input with amplitude 1 V, would an 8-bit A-law quantiser with the same dynamic range perform better?

14. A compact disc (CD) records audio signals digitally using PCM. Assume that the audio signal bandwidth is 15 kHz.

    (a) If the Nyquist samples are uniformly quantized into $L = 65,536$ levels and then binary coded, determine the number of binary digits required to encode a sample.

    (b) If the audio signal has average power of 0.1 watt and peak voltage of 1 volt, find the resulting signal-to-quantization-noise ratio (SQNR) of the uniform quantizer output.

    (c) Practical CDs sample the signal at $44,100$ samples/sec. If $L = 65,536$, determine the number of bits per second required to encode the signal.

15. The message signal $m(t)$ has a bandwidth of 10KHz, a power of 16W and a maximum amplitude of 6. It is desirable to transmit this message to a destination via a channel with 80dB attenuation and additive white noise with PSD $S_n(f) = \frac{N_0}{2} = 10^{-12} W/Hz$, and achieve a SNR at the demodulator output of at least 50dB. What is the required transmitter power and channel bandwidth if the following modulation schemes are employed? Assume ideal coherent demodulator.

    a) DSB-SC AM

b) Conventional AM with modulation index equal to 0.8. consider the modulated signal as

$$u(t) = A_c[1 + am_n(t)] \cos(2\pi f_c t),$$

$m_n(t)$ is normalized so that its max magnitude 1.

16. A DSB amplitude-modulated signal with power-spectral density as shown in Figure is corrupted with additive noise that has a power-spectral density N0/2 within the passband of the signal. The received signal-plus-noise is demodulated and low pass filtered as shown in Figure. Determine the SNR at the output of the LPF.

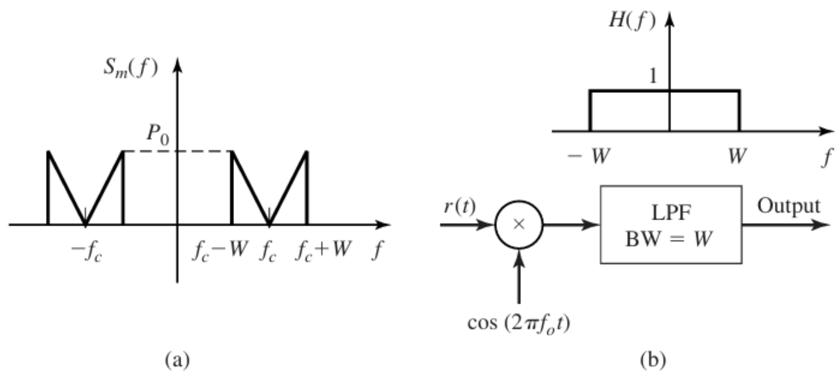

Figure 1: Q:16

17. A sinusoidal message signal whose frequency is less than 1000 , modulates the carrier $c(t) = 10^{-3}\cos(2\pi f_c t)$. The modulation scheme is conventional AM and the modulation index is 0.5. The channel noise is additive white with power spectral density of $\frac{N_0}{2} = 10^{-12} W/Hz$. At the receiver, the signal is processed as shown in Figure. The frequency response of the bandpass noise limiting filter is shown in Figure.

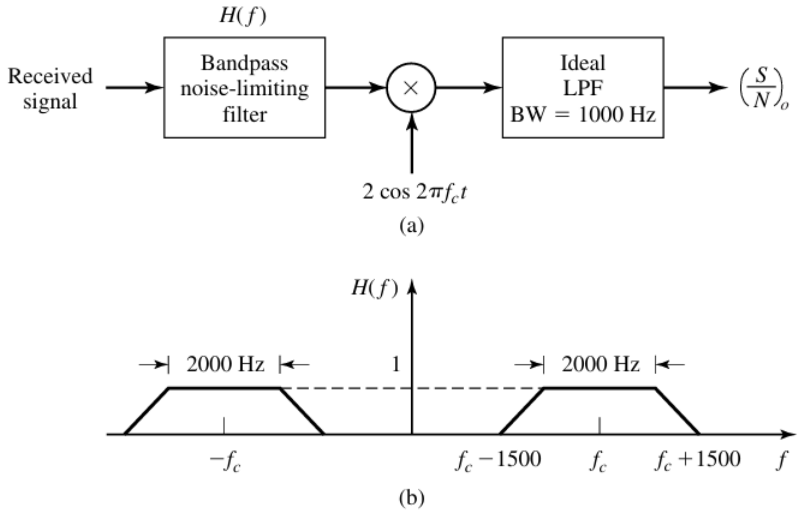

Figure 2: Q:17

1. Find the signal power and the noise power at the output of the noise-limiting filter.
2. Find the output SNR.